Project 16: Part-of-Speech Tagging: Hidden Markov Model vs Bidirectional LSTM Tagger

Phase 1 Baseline: Generative HMM and Rigorous Evaluation on UD English-EWT

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Project Repository: https://github.com/r0hansng/aml-project

Abstract*—Part-of-Speech (POS) tagging is a fundamental sequence labeling task that maps natural language sentences into grammatical category sequences. In this Phase 1 investigation, we formulate, calibrate, and evaluate a generative First-Order Hidden Markov Model (HMM) on the Universal Dependencies English Web Treebank (UD English-EWT, v2.18). We resolve three core challenges in statistical sequence modeling: structural boundary modeling via dedicated ⟨START⟩ and ⟨END⟩ context states, parameter estimation under data sparsity via Laplace add-α smoothing, and unknown-word evaluation integrity via strict separation between the empirical training vocabulary V_{original} ($19,674$ distinct types) and the model vocabulary V_{model} ($19,675$ entries including ⟨UNK⟩). Sequence decoding is executed in exact log-space via the Viterbi dynamic programming trellis. Using strict split discipline, we tune the smoothing parameter on the development split, selecting $\alpha^* = 0.10$. On the isolated official test set ($25,094$ tokens), the model achieves 87.59% overall token accuracy, 0.7944 macro-averaged F_1, and 30.98% unknown-word accuracy across $2,292$ out-of-vocabulary tokens (9.13% test OOV rate). We provide fine-grained per-tag metrics, linguistic confusion analysis, qualitative inference evaluations on film dialogues, and establish the theoretical foundation for Phase 2 Bidirectional LSTM and hybrid neural sequence labeling.

***Index Terms*—Part-of-Speech Tagging, Hidden Markov Model, Viterbi Algorithm, Add-α Smoothing, Universal Dependencies, Sequence Labeling, Out-of-Vocabulary Evaluation.**

I. Introduction

Assigning part-of-speech (POS) tags to words is an essential preprocessing step for downstream language understanding, syntactic parsing, and semantic information extraction. Under the Universal Dependencies (UD) framework [1], tokens are annotated using a cross-linguistically consistent inventory of 17 Universal POS (UPOS) categories.

Formally, given an observation sequence of tokens $W = (w_1, w_2, \ldots, w_n)$, the sequence labeling objective seeks the tag sequence $\hat{T} = (t_1, t_2, \ldots, t_n) \in S^n$ that maximizes the joint posterior probability:

$$\hat{T} = \arg\max_{T \in S^n} P(T \mid W) = \arg\max_{T \in S^n} P(W, T) \qquad (1)$$

Direct exploration of this space requires searching over $|S|^n = 17^n$ sequences, which is computationally intractable for sentences exceeding 10 words.

In Phase 1 of this project, we implement a mathematically grounded First-Order HMM baseline [4], [3] on the English Web Treebank (EWT) [2]. We introduce three core engineering solutions:

1) **Dual-Vocabulary Separation**: Enforcing strict mathematical isolation between the pure empirical training vocabulary (V_{original}) and the model parameter vocabulary (V_{model}), preventing evaluation data leakage.
2) **Laplace Add-α Smoothing**: Formulating additive smoothing on both transition and emission distributions with explicit boundary normalization [5].
3) **Log-Space Viterbi Trellis**: Formulating the dynamic programming recurrence in logarithmic space to eliminate floating-point underflow [6].

II. Related Work

Our methodology synthesizes principles from foundational sequence modeling literature:

- **Nivre et al. (2020)** [1]: Defines the canonical 17-tag UPOS standard.
- **Silveira et al. (2014)** [2]: Establishes the gold-standard English Web Treebank splits ($12,544$ train, $2,001$ dev, $2,077$ test sentences across 5 web domains) and CoNLL-U format constraints.
- **Church (1988)** [3]: Establishes the stochastic factorization of sequence tagging into local tag transitions and lexical word emissions.
- **Rabiner (1989)** [4]: Formulates HMM parameterization $\lambda = (S, V, A, B, \pi)$ and dynamic programming Viterbi decoding.
- **Brants (2000)** [5]: Proposes boundary tokens (⟨START⟩, ⟨END⟩), frequency-based lexicon pruning, and additive smoothing.
- **Manning et al. (2008)** [7]: Mandates strict train/dev/test split isolation.
- **Sokolova & Lapalme (2009)** [8]: Proves that macro-averaged F_1 and confusion matrices are essential for evaluating skewed multi-class distributions.

III. Corpus Properties and Exploratory Analysis

The UD English-EWT dataset consists of informal and formal web text. Our exploratory data analysis revealed critical statistical properties across partitions, summarized in Table I.

TABLE I
EWT Split Statistics and Out-of-Vocabulary Distribution

Split	Sentences	Tokens	Mean L	OOV Types	OOV Token %
Train	12,544	204,578	16.31	—	—
Dev	2,001	25,148	12.57	1,661	8.30%
Test	2,077	25,094	12.08	1,836	9.13%

A. Zipfian Class Imbalance

The tag frequency distribution in the training corpus is acutely skewed. Nominals (NOUN, $34,751$ tokens; 16.99%), punctuation (PUNCT, $23,710$; 11.59%), and verbs (VERB, $23,080$; 11.28%) constitute nearly 40% of the corpus. Conversely, tail categories such as interjections (INTJ, 935; 0.46%), symbols (SYM, 524; 0.26%), and unclassified/foreign tokens (X, 400; 0.20%) comprise under 1%. Consequently, global accuracy alone is an insufficient measure of model quality.

B. Lexical Ambiguity

A total of $1,973$ distinct word types in the training set appear with two or more UPOS tags. Highly frequent words exhibit substantial ambiguity (e.g., *that* occurs as SCONJ, PRON, DET, and ADV). Resolving this ambiguity requires sequence context, which the HMM provides via state transition probabilities $P(t_k \mid t_{k-1})$.

IV. Methodology and Model Architecture

A. Dual-Vocabulary Disciplinary Separation

To guarantee evaluation integrity, we enforce two separate vocabulary representations:

- V_{original}: The pure empirical training vocabulary containing all $19,674$ distinct word types observed in `train.conllu`. It strictly excludes $\langle$UNK$\rangle$.
- V_{model}: The model vocabulary consisting of words meeting the minimum frequency threshold (≥ 1) plus the special out-of-vocabulary symbol $\langle$UNK$\rangle$ ($|V_{\text{model}}| = 19,675$).

When evaluating out-of-vocabulary performance on unseen data, a token w is defined as unknown if and only if $w \notin V_{\text{original}}$.

B. HMM Generative Parameterization

Under the first-order Markov assumption and output independence, the joint probability factorizes as:

$$P(W,T) = P(t_1 \mid \langle\text{START}\rangle)P(w_1 \mid t_1) \cdot \prod_{k=2}^{n} P(t_k \mid t_{k-1})P(w_k \mid t_k) \cdot P(\langle\text{END}\rangle \mid t_n) \tag{2}$$

Transition probabilities are smoothed via add-α:

$$P(t_j \mid t_i) = \frac{C(t_i, t_j) + \alpha}{C(t_i) + \alpha \cdot |S_{\text{next}}|} \tag{3}$$

where $|S_{\text{next}}| = 17$ for $t_i = \langle$START$\rangle$, and $|S_{\text{next}}| = 18$ for $t_i \in S$ (including $\langle$END$\rangle$).

Emission probabilities are similarly smoothed:

$$P(w \mid t_i) = \frac{C(t_i, w) + \alpha}{C(t_i) + \alpha \cdot |V_{\text{model}}|} \tag{4}$$

where $|V_{\text{model}}| = 19,675$.

C. Log-Space Viterbi Trellis Algorithm

To prevent numerical underflow, the dynamic programming recurrence is evaluated strictly in log-space. The complete decoding procedure is specified in Algorithm 1.

Algorithm 1 Log-Space Viterbi Trellis Decoding

Require: Sequence $W = (w_1, \ldots, w_n)$, state set S, log transitions $\log A$, log emissions $\log B$.
Ensure: Optimal state sequence $\hat{T} = (\hat{t}_1, \ldots, \hat{t}_n)$.

Initialization ($t = 1$)**:**
for all $j \in S$ **do**
  $w'_1 \leftarrow w_1$ if $w_1 \in V_{\text{model}}$ else $\langle$UNK$\rangle$
  $v_1(j) \leftarrow \log A[\langle\text{START}\rangle, j] + \log B[j, w'_1]$
end for
Recursion ($t = 2, \ldots, n$)**:**
for $t = 2$ **to** n **do**
  $w'_t \leftarrow w_t$ if $w_t \in V_{\text{model}}$ else $\langle$UNK$\rangle$
  for all $j \in S$ **do**
    $bp_t(j) \leftarrow \arg\max_{i \in S} (v_{t-1}(i) + \log A[i, j])$
    $v_t(j) \leftarrow \max_{i \in S} (v_{t-1}(i) + \log A[i, j]) + \log B[j, w'_t]$
  end for
end for
Termination:
$\hat{t}_n \leftarrow \arg\max_{j \in S} (v_n(j) + \log A[j, \langle\text{END}\rangle])$
Backtracking:
for $t = n - 1$ 1 **do**
  $\hat{t}_t \leftarrow bp_{t+1}(\hat{t}_{t+1})$
end for
return $\hat{T} = (\hat{t}_1, \ldots, \hat{t}_n)$

This formulation guarantees exact global MAP decoding in $\mathcal{O}(T \cdot |S|^2)$ time and $\mathcal{O}(T \cdot |S|)$ auxiliary space.

V. Experiments and Results

A. Hyperparameter Tuning on Development Split

Following Manning et al. [7], the smoothing parameter $\alpha \in \{0.01, 0.1, 1.0\}$ was tuned strictly on the development set:

TABLE II
Hyperparameter Tuning Results on Development Split

Smoothing α	Token Acc (%)	Macro F_1	Unknown Acc (%)
$\alpha = 0.01$	87.13%	0.7716	24.92%
$\alpha = 0.10$ **(Selected)**	**87.41%**	**0.7817**	**29.52%**
$\alpha = 1.00$	84.52%	0.7491	33.45%

The optimal value $\alpha^* = 0.10$ maximizes both Token Accuracy (87.41%) and Macro F_1 (0.7817). Standard Laplace

smoothing ($\alpha = 1.0$) overly diffuses probability mass over the large vocabulary ($19,675$ items), causing a sharp 2.89% drop in token accuracy.

B. Final Test Set Performance

Evaluating the frozen optimal model ($\alpha^* = 0.10$) once on the test partition ($2,077$ sentences, $25,094$ tokens) yields:

- **Overall Token Accuracy**: $\mathbf{87.59\%}$
- **Macro-Averaged** F_1: $\mathbf{0.7944}$
- **Unknown-Word Accuracy**: $\mathbf{30.98\%}$ ($670/2,292$ correct)
- **Unknown Tokens Evaluated**: $2,292$ (9.13% OOV rate)

The complete per-tag performance breakdown is reported in Table III.

TABLE III
DETAILED PER-TAG CLASSIFICATION REPORT ON OFFICIAL TEST SET

Tag	Precision	Recall	F_1-Score	Support
CCONJ	0.97	0.98	**0.98**	736
PRON	0.93	0.97	**0.95**	2,164
PUNCT	0.93	0.98	**0.95**	3,096
DET	0.91	0.97	**0.94**	1,897
AUX	0.89	0.98	**0.93**	1,543
ADP	0.89	0.94	**0.92**	2,025
PART	0.86	0.93	**0.90**	649
VERB	0.91	0.86	**0.89**	2,605
NOUN	0.92	0.82	**0.87**	4,123
ADJ	0.87	0.84	**0.85**	1,788
ADV	0.81	0.85	**0.83**	1,191
NUM	0.78	0.69	**0.73**	542
PROPN	0.83	0.61	**0.70**	2,075
SCONJ	0.70	0.69	**0.69**	384
INTJ	0.58	0.79	**0.66**	121
SYM	0.50	0.80	**0.62**	113
X	0.05	0.52	**0.10**	42
Macro Avg	**0.78**	**0.84**	**0.79**	**25,094**

C. Error Analysis and Confusion Dynamics

Inspection of the test set confusion matrix reveals dominant linguistic failure modes:

1) **NOUN** $\rightarrow$ **PROPN** (161 errors) and **PROPN** $\rightarrow$ **NOUN** (153 errors): Web text exhibits capitalization noise (all-caps titles, uncapitalized internet slang, product names).
2) **PROPN** $\rightarrow$ **X** (136 errors): Rare named entities, foreign names, and web handles are misclassified into the catch-all category.
3) **PROPN** $\rightarrow$ **PUNCT** (118 errors): Acronyms containing internal periods and web tokens containing punctuation.
4) **NOUN** $\rightarrow$ **X** (112 errors): Informal typos and colloquial web jargon unobserved in standard training text.

VI. QUALITATIVE DEMONSTRATIONS

We evaluated qualitative inference on stylized dialogue from David Fincher's *The Social Network* (2010), reported in Table IV.

The token *inventors* is an out-of-vocabulary word ($w \notin V_{\text{original}}$). Despite zero lexical counts, the HMM correctly resolves it as NOUN due to the strong transition prior from the preceding determiner *the* ($P(\texttt{NOUN} \mid \texttt{DET}) = 0.61$).

TABLE IV
QUALITATIVE EVALUATION ON FINCHER DIALOGUE SEQUENCE

Token	Pred	Status	Syntactic Note
If	SCONJ	Known	Subordinate conditional marker.
you	PRON	Known	2nd person subject pronoun.
guys	NOUN	Known	Plural colloquial nominal.
were	AUX	Known	Subjunctive copular auxiliary.
the	DET	Known	Definite article.
inventors	**NOUN**	**UNK**	**True OOV**: correctly resolved via $P(\texttt{NOUN} \mid \texttt{DET})$.
of	ADP	Known	Preposition introducing PP.
Facebook	PROPN	Known	Proper named entity noun.
,	PUNCT	Known	Clause delimiter.
you	PRON	Known	Matrix subject pronoun.
would	AUX	Known	Modal auxiliary verb.
have	AUX	Known	Perfect auxiliary verb.
invented	VERB	Known	Past participle main verb.
Facebook	PROPN	Known	Direct object named entity.
.	PUNCT	Known	Sentence-final delimiter.

VII. ROADMAP TO PHASE 2: BILSTM AND HYBRID TAGGING

While the first-order HMM performs strongly on closed-class words ($F_1 > 94\%$), its 30.98% unknown-word accuracy highlights fundamental Markovian limitations:

1) **Finite Context Horizon**: $P(t_k \mid t_{k-1})$ cannot capture long-range grammatical agreements across subordinate clauses.
2) **Emission Independence**: Unigram word emissions discard subword morphology (e.g., *-ing*, *-tion*, *-ed*).
3) **Coarse OOV Pooling**: All unknown words share identical emission probabilities.

In Phase 2, we will implement a Bidirectional LSTM (BiLSTM) with pre-trained word embeddings and character-level subword representations. Furthermore, we will investigate a **Hybrid BiLSTM-CRF** architecture to combine dense continuous representations with structured transition decoding.

VIII. TEAM DIVISION OF RESPONSIBILITIES

Individual technical contributions for Phase 1 are summarized below:

- **Rohan Singh** (2401010391): Project build orchestration, automated test suite (31 unit/integration tests), literature review, and technical report authoring.
- **Aniket Pathak** (2401020088): First-order HMM formulation with add-α smoothing, log-space Viterbi decoder, and sequence evaluation metrics.
- **Pushpendra Singh Parihar** (2401010361): Corpus preprocessing, sentence boundary tagging, vocabulary pruning, and dual-vocabulary evaluation protocol.
- **Kapil Karan Mathur** (2401010209): Hyperparameter grid tuning (α search), test evaluation scripts, interactive notebook, and diagnostic visualizations.
- **Agrim Kumar Malhotra** (2401010045): CoNLL-U parser with multi-word token handling, dataset validation suite, and exploratory data analysis.

IX. Conclusion

We have implemented, tuned, and evaluated a First-Order HMM POS tagger on UD English-EWT. By maintaining strict vocabulary separation, deriving smoothed boundary distributions, and decoding in log-space, the model achieves an 87.59% test token accuracy and 0.7944 macro F_1. All code, data pipelines, and test suites (31 automated unit tests) are reproducible and hosted in the public repository.

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